Outline

1. Maximum and Minimum Values
2. The Mean Value Theorem
3. Derivative Tests
4. **Curve Sketching: a self-study**
5. Applied Optimization Problems
6. L'Hospital's Rule
7. Newton's Method
8. Antiderivatives

In this section, we study several applied optimization problems, which ask for the best, or optimal, value of a given function.

More precisely, the following questions can be asked:

- What are the dimensions of a rectangle with fixed perimeter having maximum area?
- What are the dimensions for the least expensive cylindrical can of a given volume?
- How many items should be produced for the most profitable production run?
- etc.

Our aim is to use derivatives to solve a variety of optimization problems in mathematics, physics, economics, and business.

To solve these applied optimization problems, we should follow the following steps:

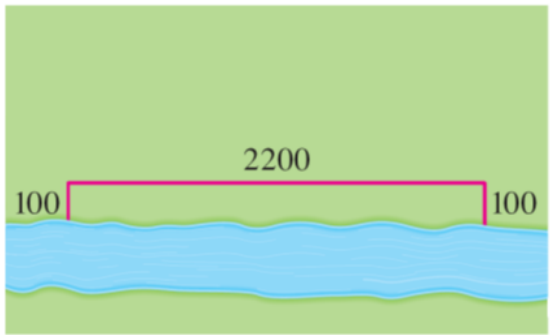
- **Understand the problem**: *Read the problem carefully, until you understand it. What are the given quantities? What are the given conditions? What is the unknown to be optimized?*
- **Draw a diagram**: *Draw a diagram. Label any part that may be important to the problem.*
- **Introduce variables**: *Assign a symbol to the quantity that is to be maximized or minimized (let's call it Q for now). Also select symbols* $(a, b, c, \ldots, x, y)$ *for other unknown quantities and label the diagram with these symbols (Use initials as suggestive symbols, for example, A for area, h for height, t for time).*

- **Working steps**:
  - *Express Q in terms of some of the other symbols. If Q has been expressed as a function of more than one variable, use the given information to find relationships (in the form of equations) among these variables. Then use these equations to eliminate all but one of the variables in the expression for Q.*
  - *Q is then expressed as a function of one variable x, say, $Q = f(x)$. Write the domain of this function in the given context.*
  - *Find the absolute maximum or minimum value of $f(x)$. In particular, if the domain of $f(x)$ is a closed interval, then the Closed Interval Method can be used.*

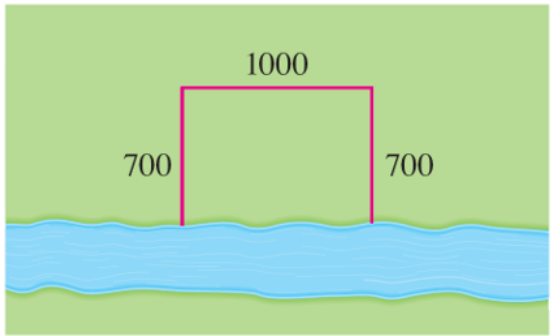
Example

A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

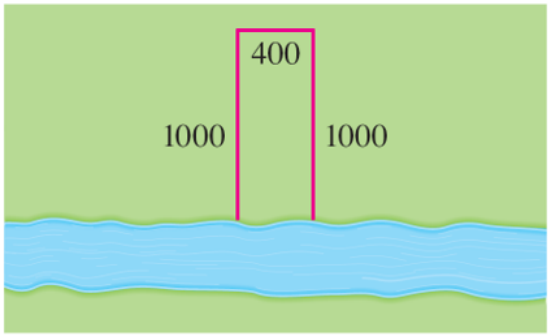
Solution. In order to get a feeling for what is happening in this problem, let's experiment with some specific cases. Figure below shows three possible ways of laying out the $2,400$ ft of fencing.



Area $= 100 \cdot 2200 = 220{,}000$ ft^2



Area $= 700 \cdot 1000 = 700{,}000$ ft^2



Area $= 1000 \cdot 400 = 400{,}000$ ft^2

Let x and y be the depth and width of the rectangle (in feet). Then the area A can be expressed in terms of x and y:

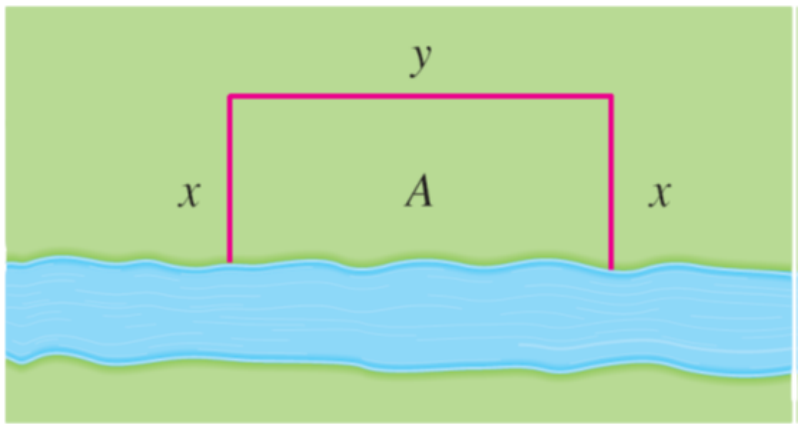
$$A = xy.$$

We hope to express A as a function of just one variable, so we eliminate y by expressing it in terms of x. That is,

$$2x + y = 2,400 \implies y = 2400 - 2x.$$

Then we can have the expression for the area A as following

$$A = xy = x(2,400 - 2x) = 2,400x - 2x^2.$$



Conditions for x: $0 \leq x \leq 1,200$ (why?). Then the function to maximize is

$$A = 2,400x - 2x^2, \quad 0 \leq x \leq 1.200,$$

whose critical points are

$$A'(x) = 0 \iff 2,400 - 4x = 0 \iff x = 600.$$

The maximum value of A must occur either at critical points or at an endpoints of the interval. So we compare

$$A(0) = 0, \quad A(600) = 720,000, \quad A(1,200) = 0,$$

and hence the Closed Interval Method gives the maximum value as $A(600) = 720,000$. Hence, $y = 2,400 - 2 * 600 = 1,200$.

So the rectangular field should be 600 ft deep and $1,200$ ft wide.

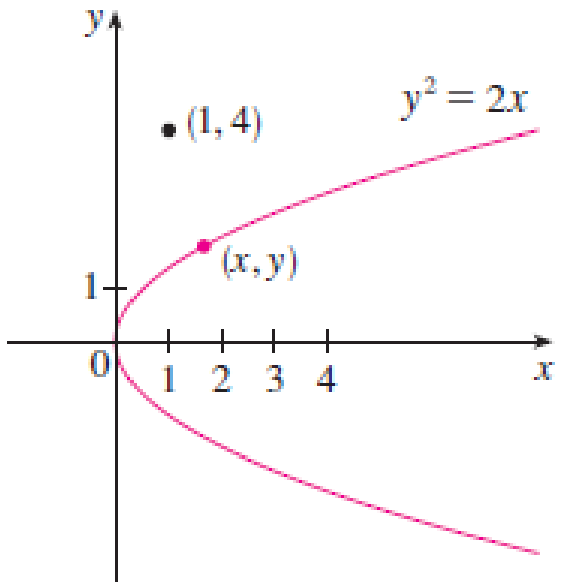
Example

Find the points on parabola $y^2 = 2x$ that are closest to the point $(1, 4)$.

Solution. The distance between the point (1,4) and the point (x,y) is

$$d = \sqrt{(\underbrace{x}_{\frac{1}{2}y^2} - 1)^2 + (y-4)^2}, \quad \text{or} \quad d(y) = \sqrt{\left(\frac{1}{2}y^2 - 1\right)^2 + (y-4)^2}.$$

The domain of $d(y)$ is $(-\infty, \infty)$.



Instead of minimizing $d(y)$, we minimize its square:

$$f(y) = d^2 = \left(\frac{1}{2}y^2 - 1\right)^2 + (y-4)^2$$

(because the $\min(d)$ occurs at the same points as the $\min(d^2)$, and d^2 is easier to work with).

We have

$$f'(y) = 2\left(\frac{1}{2}y^2 - 1\right)y + 2(y-4) = y^3 - 8 = 0 \iff y = 2.$$

Observe that

$$\begin{cases} f'(y) < 0 & \text{when } y < 2 \\ f'(y) > 0 & \text{when } y > 2, \end{cases}$$

and so the absolute minimum occurs when $y = 2$.

The corresponding value of x is $x = \frac{1}{2}y^2 = 2$.

Thus the point $(2, 2)$ on $y^2 = 2x$ is closest to $(1, 4)$ (and the distance between these points is $d = \sqrt{f(2)} = \sqrt{5}$).

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Let us have a look at the limit

$$\lim_{x \to 1} \frac{\ln x}{x - 1}.$$

The limit is not obvious, because both numerator and denominator approach 0 and $\frac{0}{0}$ is not defined.

In general, if we have a limit of the form

$$\lim_{x \to a} \frac{f(x)}{g(x)},$$

where both $f(x) \to 0$ and $g(x) \to 0$ as $x \to a$, the limit may or may not exist. This form is called an indeterminate form of type $\frac{0}{0}$.

We already know the following:

- For rational functions, we can cancel common factors:

$$\lim_{x\to 1}\frac{x^2-x}{x^2-1}=\lim_{x\to 1}\frac{x(x-1)}{(x+1)(x-1)}=\lim_{x\to 1}\frac{x}{x+1}=\frac{1}{2}.$$

- For the trigonometrical limit

$$\lim_{x\to 0}\frac{\sin x}{x}=1.$$

But both do not work for the limit $\lim_{x\to 1}\dfrac{\ln x}{x-1}$.

Moreover, when we calculate the horizontal asymptote for $F(x) = \dfrac{\ln x}{x-1}$, we need to know

$$\lim_{x\to\infty} \frac{\ln x}{x-1}.$$

It isn't obvious how to evaluate this limit, because both numerator and denominator become ∞ as $x \to \infty$.

In general, if we have a limit of the form

$$\lim_{x\to a} \frac{f(x)}{g(x)},$$

where both $f(x) \to \infty$ (or $-\infty$) and $g(x) \to \infty$ (or $-\infty$), the limit may or may not exist. This form is called an indeterminate form of type $\dfrac{\infty}{\infty}$.

L'Hospital's Rule applies to these indeterminate forms.

L'Hospital's Rule

Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \to a} f(x) = 0 \quad \text{and} \quad \lim_{x \to a} g(x) = 0,$$

or that

$$\lim_{x \to a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \to a} g(x) = \pm\infty.$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$). Then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

if the limit on the right-hand side exists (or is ∞ or $-\infty$).

Note.

- L'Hospital's Rule says that the limit of a quotient of functions is equal to the limit of the quotient of their derivatives. It is especially important to verify the conditions regarding the limits of $f(x)$ and $g(x)$ before using L'Hospital's Rule.
- L'Hospital's Rule is valid for one-sided limits and for limits at infinity or negative infinity. That is, "$x \to a$" can be replaced by any of the symbols $x \to a^+$, $x \to a^-$, $x \to \infty$, or $x \to -\infty$.

Example

Find

$$\lim_{x\to 1} \frac{\ln x}{x-1}.$$

Solution. We have

$$\lim_{x\to 1} \ln x = \ln 1 = 0 \quad \text{and} \quad \lim_{x\to 1} (x-1) = 0.$$

Therefore, the limit is an indeterminate form of type $\frac{0}{0}$.

Applying L'Hospital's Rule, we get

$$\lim_{x\to 1} \frac{\ln x}{x-1} = \lim_{x\to 1} \frac{(\ln x)'}{(x-1)'} = \lim_{x\to 1} \frac{1/x}{1} = \lim_{x\to 1} \frac{1}{x} = 1.$$

If $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = \infty$ (or $-\infty$), then it isn't clear what the value of $\lim_{x\to a}[f(x)g(x)]$, if any, will be.

This kind of limit is called an indeterminate form of type $0 \cdot \infty$.

We can deal with it by writing the product fg as a quotient:

$$fg = \frac{f}{1/g} \text{ or } fg = \frac{g}{1/f}.$$

This converts the given limit into an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$, so that we can use L'Hospital's Rule.

Example

Evaluate

$$\lim_{x \to 0^+} x \ln x.$$

Solution. The given limit is indeterminate. Because as $x \to 0^+$, the first factor ($x \to 0^+$), while the second factor ($\ln x \to \infty$).

Since $x = \dfrac{1}{(1/x)}$, we have $1/x \to \infty$ as $x \to 0^+$. Use L'Hospital's Rule

$$\lim_{x \to 0^+} x \ln x = \lim_{x \to 0^+} \frac{\ln x}{1/x} = \lim_{x \to 0^+} \frac{1/x}{-1/x^2} = \lim_{x \to 0^+} (-x) = 0$$

Note. Another possible way is to write $\lim_{x\to 0^+} x \ln x = \lim_{x\to 0^+} \dfrac{x}{1/\ln x}$. However, if we apply L'Hospital's Rule, we get a more complicated expression than the one we started with.

In general, when we rewrite an indeterminate product, we try to choose the option that leads to the simpler limit.

If $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = \infty$, then the limit

$$\lim_{x\to a} [f(x) - g(x)]$$

is called an indeterminate form of type $\infty - \infty$. Again there is a contest between $f(x)$ and $g(x)$.

We try to convert the difference into a quotient (for instance, by using a common denominator, or rationalization, or factoring out a common factor), so that we have an indeterminate form of type $\dfrac{0}{0}$ or $\dfrac{\infty}{\infty}$.

Example

Compute

$$\lim_{x\to(\pi/2)^-}(\sec x - \tan x).$$

Solution. Since $\sec x \to \infty$, $\tan x \to \infty$ as $x \to (\pi/2)^-$, the limit is indeterminate.

Use a common denominator,

$$\lim_{x\to(\pi/2)^-}(\sec x - \tan x) = \lim_{x\to(\pi/2)^-}\left(\frac{1}{\cos x} - \frac{\sin x}{\cos x}\right)$$

$$= \lim_{x\to(\pi/2)^-}\frac{1-\sin x}{\cos x} = \lim_{x\to(\pi/2)^-}\frac{-\cos x}{-\sin x} = 0.$$

Note that the use of L'Hospital's Rule is justified, because $1 - \sin x \to 0$ and $\cos x \to 0$ as $x \to (\pi/2)^-$.

Several indeterminate forms arise from the limit

$$\lim_{x\to a}[f(x)]^{g(x)}.$$

1. $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$, type 0^0
2. $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = 0$, type ∞^0
3. $\lim_{x\to a} f(x) = 1$ and $\lim_{x\to a} g(x) = \pm\infty$, type 1^∞

These three cases can be treated either by taking the natural logarithm:

$$y = [f(x)]^{g(x)} \to \ln y = g(x)\ln f(x),$$

or by writing the function as an exponential:

$$y = [f(x)]^{g(x)} \to y = e^{\ln y} = e^{g(x)\ln f(x)}.$$

In either method, we have the indeterminate product $g(x)\ln f(x)$, which is of type $0\cdot\infty$.

Example

Calculate

$$\lim_{x\to 0^+}(1+\sin 4x)^{\cot x}.$$

Solution. As $x \to 0^+$, we have $1+\sin 4x \to 1$ and $\cot x \to \infty$, so the given limit is indeterminate (type 1^∞). Let $y = (1+\sin 4x)^{\cot x}$, we have

$$\ln y = \cot x \ln(1+\sin 4x) = \frac{\ln(1+\sin 4x)}{\tan x}.$$

Use L'Hospital's Rule,

$$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} \frac{\ln(1+\sin 4x)}{\tan x} = \lim_{x\to 0^+} \frac{\frac{4\cos 4x}{1+\sin 4x}}{\sec^2 x} = 4.$$

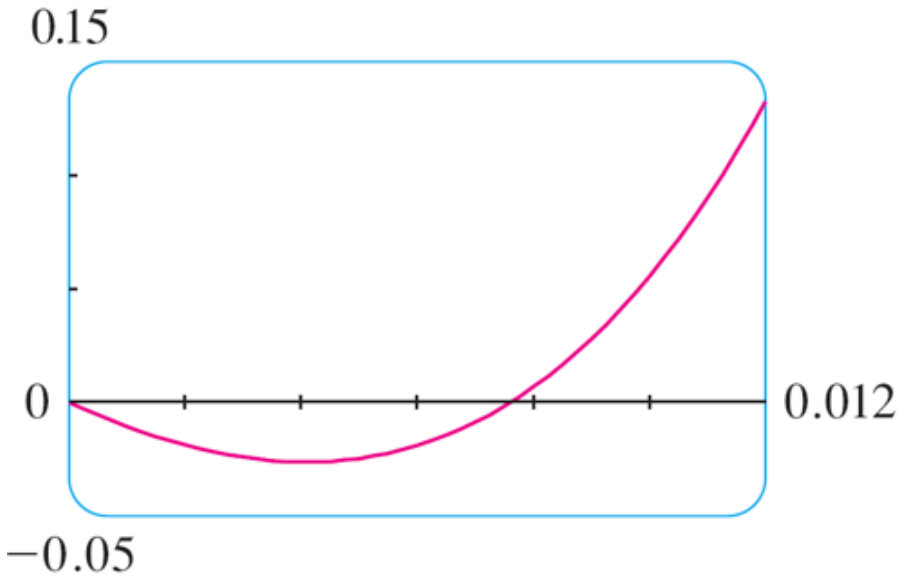
To find limit of y, we use the fact that $y = e^{\ln y}$:

$$\lim_{x\to 0^+}(1+\sin 4x)^{\cot x} = \lim_{x\to 0^+} y = \lim_{x\to 0^+} e^{\ln y} = e^4.$$

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Let's consider the equation $48x(1+x)^{60} - (1+x)^{60} + 1 = 0$, which we can find an approximate solution by using a graphing device.

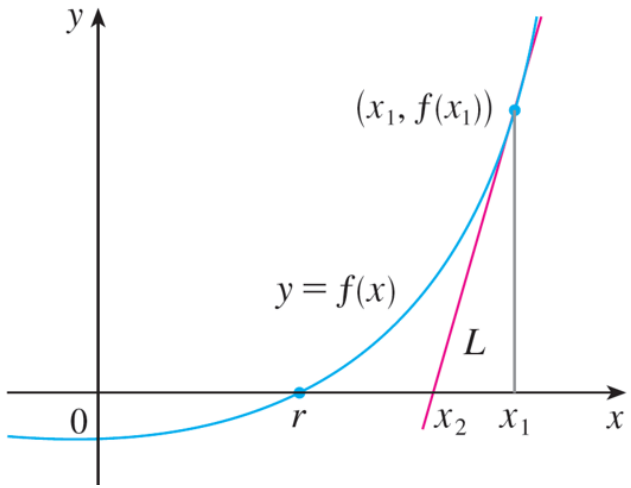


We see that besides the solution $x = 0$, there is another one between 0.007 and 0.008. Zooming in shows that the root is approximately 0.0076. If we need more accuracy we could zoom in repeatedly.

• Newton's method

Question. How can we find such approximate solutions?

- Consider the tangent line L to the curve $y = f(x)$ at the point $(x_1, f(x_1))$ and look at the x-intercept of L, labeled x_2.



The idea behind Newton's method is that the tangent line is close to the curve and so its x-intercept, x_2, is close to the root r.

Newton's Method

Moreover, x_2 can be easily calculated from tangent line

$$y - f(x_1) = f'(x_1)(x - x_1),$$

by setting $y = 0$. Namely,

$$0 - f(x_1) = f'(x_1)(x_2 - x_1) \implies x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}, \quad \text{if} \quad f'(x_1) \neq 0.$$

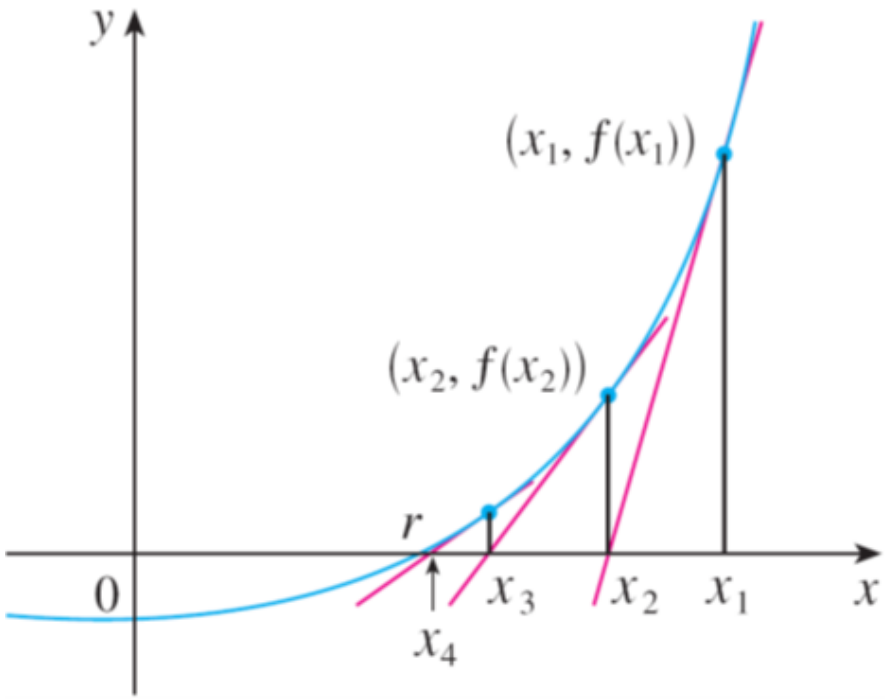
We use x_2 as a second approximation to r.

- Next we repeat this procedure with x_1 replaced by the second approximation x_2, using the tangent line at $(x_2, f(x_2))$.

This gives a third approximation:

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}, \quad \text{if} \quad f'(x_2) \neq 0.$$

- If we keep repeating this process, we obtain a sequence of approximations $x_1, x_2, x_3, x_4, \ldots$ as shown in Figure.



In general, if the n-th approximation is x_n and $f'(x_n) \neq 0$, then the next approximation is given by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

If the numbers x_n become closer and closer to r as n becomes large, then we say that the sequence converges to r and we write

$$\lim_{n \to \infty} x_n = r.$$

Remark. It should be noted that the Newton's method may fail if an improper initial approximation x_1 is used.

In this case, we need to find another initial value.

Example

Find, correct to six decimal places, the root of the equation $\cos x = x$.

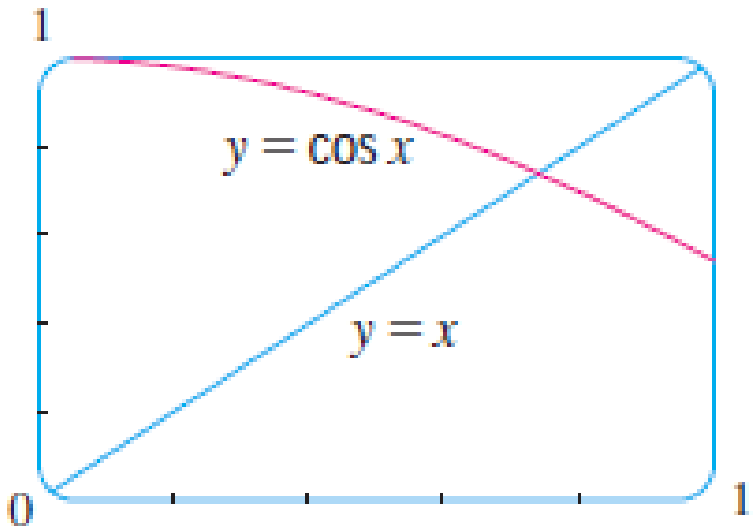
Solution. We first rewrite the equation in standard form:

$$\cos x - x = 0.$$

Therefore we let $f(x) = \cos x - x$. Then $f'(x) = -\sin x - 1$, and hence Newton's equation becomes

$$x_{n+1} = x_n - \frac{\cos x_n - x_n}{-\sin x_n - 1} = x_n + \frac{\cos x_n - x_n}{\sin x_n + 1} \quad (n \geq 1).$$

In order to guess a suitable value for x_1, we sketch the graphs of $y = \cos x$ and $y = x$ as shown in Figure.



It appears that they intersect at a point whose x-coordinate is somewhat less than 1, so let's take $x_1 = 1$ as a convenient first approximation.

Then, remembering to put our calculator in radian mode, we get

$$x_2 \approx 0.75036387$$
$$x_3 \approx 0.73911289$$
$$x_4 \approx 0.73908513$$
$$x_5 \approx 0.73908513$$

Since x_4 and x_5 agree to more than six decimal places (eight, in fact), we conclude that the root of the equation, correct to six decimal places, is 0.739085.

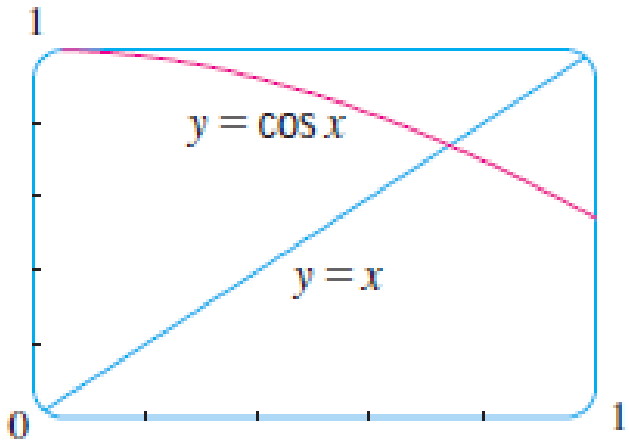
Remark. Instead of using the rough sketch to get a starting approximation for Newton's method as we did in the example above, we could have used the more accurate graph that a calculator or computer provides.

Figure below suggests that we use $x_1 = 0.75$ as the initial approximation. Then Newton's method gives

$$x_2 \approx 0.73911114$$

$$x_3 \approx 0.73908513$$

$$x_4 \approx 0.73908513$$



and so we obtain the same answer as before, but with one step fewer.

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We have studied how to find the derivative of a function and how to use it to solve a wide range of problems.

However, there are many problems that require us to recover a function from it known derivative (for instance, from its rate of change).

We now study this topic.

Definition. A function F is called an antiderivative of a function on some interval I if

$$F'(x) = f(x), \quad \text{for all } x \in I.$$

For example, consider the function $f(x) = x^2$. It is not difficult to find an antiderivative of $f(x)$: if $F(x) = \frac{1}{3}x^3$, then $F'(x) = x^2 = f(x)$.

But we can notice that the function $G(x) = \frac{1}{3}x^3 + 100$ also satisfies $G'(x) = x^2$. Therefore both $F(x)$ and $G(x)$ are antiderivatives of $f(x)$.

In fact, any function of the form $H(x) = \frac{1}{3}x^3 + C$, where C is an arbitrary constant, is an antiderivative of $f(x)$.

We have the following important result.

Theorem

If $F(x)$ is an antiderivative of $f(x)$ on an interval I, then the most general antiderivative of f on I is

$$F(x) + C,$$

where C is an arbitrary constant.

• Table of antiderivative formulas

Function	General antiderivative
1. x^n	$\frac{1}{n+1}x^{n+1} + C, \quad n \neq -1$
2. $\sin kx$	$-\frac{1}{k}\cos kx + C$
3. $\cos kx$	$\frac{1}{k}\sin kx + C$
4. $\sec^2 kx$	$\frac{1}{k}\tan kx + C$
5. $\csc^2 kx$	$-\frac{1}{k}\cot kx + C$
6. $\sec kx \tan kx$	$\frac{1}{k}\sec kx + C$
7. $\csc kx \cot kx$	$-\frac{1}{k}\csc kx + C$

Each formula in the table is true because the derivative of the function in the right column appears in the left column.

• Antiderivative linearity rules

Other derivative rules also lead to corresponding antiderivatives rules. In particular, we can add and subtract antiderivatives and multiply them by constants.

Thus we have the following table.

		Function	General antiderivative
1.	*Constant Multiple Rule:*	$kf(x)$	$kF(x) + C$, k a constant
2.	*Negative Rule:*	$-f(x)$	$-F(x) + C$,
3.	*Sum or Difference Rule:*	$f(x) \pm g(x)$	$F(x) \pm G(x) + C$

Antiderivatives are particularly useful in analyzing the motion of an object moving in a straight line. We know if the object has position function $s(t)$, then the velocity function is $v(t) = s'(t)$.

This means that the position function is an antiderivative of the velocity function.

Likewise, since the acceleration function is $a(t) = v'(t)$, the velocity function is an antiderivative of the acceleration.

So if the acceleration and the initial values $s(0)$ and $v(0)$ are known, then the position function can be found by antiderivative rules twice.

Example

A ball is thrown upward with a speed of 15 m/s from the edge of a cliff 140 m above the ground. Find its height above the ground after t sec. When does it reach its maximum height? When does it hit the ground?

Solution. The ball moves upward, so we choose the positive direction to be upward.

At time t the distance above the ground is $s(t)$ and the velocity $v(t)$ is decreasing. Therefore, the acceleration must be negative and we have

$$a(t) = \frac{dv}{dt} = -9.8$$

Taking antiderivatives, we have

$$v(t) = -9.8t + C, \quad \text{where } C \text{ is an arbitrary constant.}$$

Furthermore, since $v(0) = 15$, we have $15 = 0 + C$, that is $C = 15$. Thus $v(t) = -9.8t + 15$.

The maximum height is reached when $v(t) = 0$, that is, after $15/9.8 \approx 1.53$ seconds.

Since $s'(t) = v(t)$, we use antiderivative rules again to obtain

$$s(t) = -4.9t^2 + 15t + D, , \quad \text{where } D \text{ is an arbitrary constant.}$$

Using the condition $s(0) = 140$, we have $140 = 0 + D$, that is $D = 140$. Thus $s(t) = -4.9t^2 + 15t + 140$.

The expression for $s(t)$ is valid until the ball hits the ground. This happens when $s(t) = 0$, that is, when

$$4.9t^2 - 15t - 140 = 0.$$

Solving this quadratic equation, we obtain

$$t = \frac{15 \pm \sqrt{2969}}{9.8} \approx 7.1 \text{ sec} \quad (\textit{we ignore the negative value})$$

• Indefinite integrals

In mathematics, there is a special symbol for denoting the collection of all antiderivatives of a function.

Definition. The collection of all antiderivatives of f is called the indefinite integral of f w.r.t. x, and is denoted by

$$\int f(x)\,dx.$$

- The symbol $\int$ is an integral sign
- The function f is the integrand of the integral
- x is the variable of integration.

Note. After the integral sign in the notation above, the integrand function is always followed by a differential to indicate the variable of integration.

By this, we can write

- $\int dx = x + C$,
- $\int \cos x \, dx = \sin x + C$,
- etc.

END OF CHAPTER 3